

Rethinking compilers with live coding

Tomas Petricek, Alan Turing Institute + fsharpWorks
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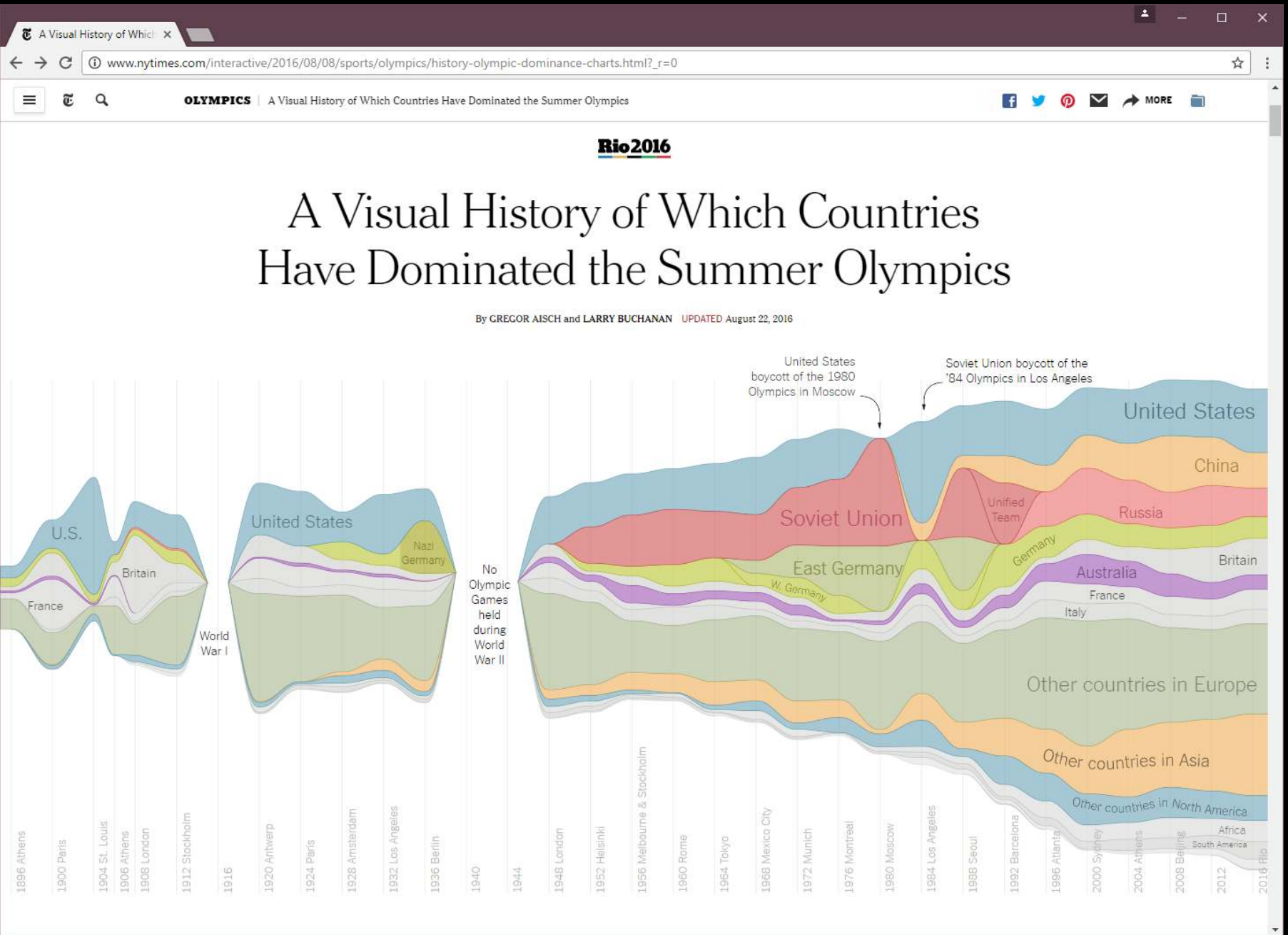
29

NOT #INFOR THIS?

Welcome to the post-fact world

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Can the result be reproduced?

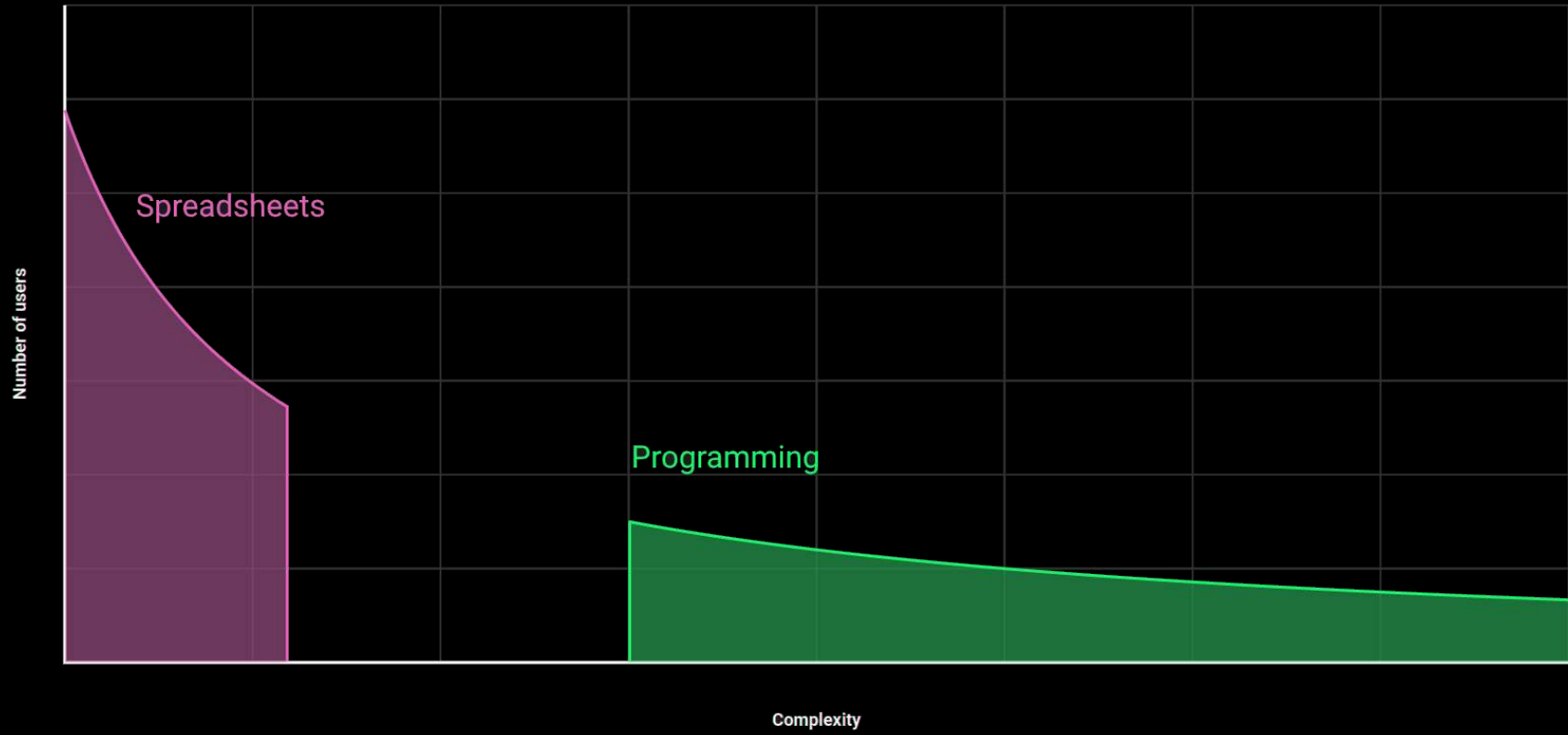
Is the visualization misleading?

Can the reader explore further?



The Gamma

Making data science easier





Book1 - Excel Tomas Petricek

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D4

	A	B	C	D	E	F	G	H	I
1									
2									
3		Cost	Rate	Converted					
4		149	0.63						
5		99	1.23						
6		14	0.63						
7		99	1						
8		249	1.23						
9									
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13									

Sheet1

Ready 120%



Book1 - Excel

Tomas Petricek



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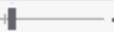
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Sheet1





Ready



120%

DEMO

Live previews in TheGamma

TEXTBOOK COMPILERS

Textbook compiler is batch based

Textbook compiler builds & transforms trees

Textbook compiler rejects bad programs

Example - live image processing

```
let pope = image.load("pope.png")  
let shadow = image.load("shadow.png")  
  
shadow.greyscale().blur(5)  
  .combine(pope, 50)
```

SKETCH

Transforming abstract syntax trees

DEMO

Writing a batch evaluator

LIVE COMPILERS

Live compiler caches earlier results

Live compiler builds dependency graphs

Live compiler accepts bad programs

SKETCH

Building a dependency graph

DEMO

Writing a graph-based evaluator

SUMMARY

Check out **TheGamma** project

And help make facts great again!

www.thegamma.net

gamma.turing.ac.uk

F# language

www.fsharp.org

Pragmatic functional-
first programming

Fable compiler

www.fable.io

F# to JavaScript compiler
for the 21st century

Dependency graphs over syntax trees

Reusing nodes enables caching

Good luck writing a decent parser :-)

Now you know how to write
your own live coding environment!

Questions?

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